

Model Explanations

research question:

Can we predict whether a U.S. adult earns at most \$50,000 per year based on their education and employment characteristics?

The 4 Concepts Simply Explained

1. Support (The Quantity):

- **What it is:** The exact number of people from the test set who *actually* belong in that category.

2. Precision (How Reliable is the Guess?):

- **What it is:** When the model says "*This person falls into category X*", how often is the model actually **right**? A score of 0.76 means that if the model claims 100 times that someone earns $\leq 50K$, it is correct in 76 cases and guessing wrong in 24 cases.

3. Recall (How Much Did We Catch?):

- **What it is:** What percentage of the *real* target group was the model able to fish out? If there are 100 rich people in the room and your model only catches 20, your recall is low (20%). If you catch all of them, it is 1.00 (100%).

4. F1-Score (The Total Score):

- **What it is:** A handy summary (an average) of Precision and Recall. It runs from 0.0 to 1.0. The closer it is to 1.0, the better balanced the model is for that group.

Model	Best Hyperparameters	Cross-Validation Score (Mean $\pm$ Std)	Test Precision (>50K)	Test Recall (>50K)	Test F1-score (>50K)
Dummy Classifier (Baseline)	None (Strategy: Most Frequent)	0.759 $\pm$ 0.001	0.00	0.00	0.00

Model	Best Hyperparameters	Cross-Validation Score (Mean $\pm$ Std)	Test Precision (>50K)	Test Recall (>50K)	Test F1-score (>50K)
K-Nearest Neighbors (KNN)	n_neighbors: 15, weights: 'uniform'	0.835 $\pm$ 0.004	0.67	0.60	0.64
Logistic Regression	C: 1.0, solver: 'lbfgs'	0.839 $\pm$ 0.003	0.70	0.58	0.64
Random Forest	n_estimators: 100, max_depth: 15	0.852 $\pm$ 0.003	0.69	0.62	0.65

1. The Confusion Matrix of Dummy Classifier (Baseline)

The matrix illustrates how the model's predictions align with reality:

- **Top-left (7422):** These are people who actually earned $\leq 50K$, and the model correctly predicted this.
- **Top-right (0):** These are people who earned $\leq 50K$, but the model mistakenly classified them as wealthy ($> 50K$).
- **Bottom-left (2336):** These are people who actually earned $> 50K$, but the model incorrectly predicted they earned $\leq 50K$.
- **Bottom-right (0):** These are people who actually earned $> 50K$, and the model correctly predicted this.

2. The Classification Report

The Group $\leq 50K$ (People earning \$50,000 or less)

- **Support = 7422:** The total number of people in this actual category.
- **Precision = 0.76 (76%):** When the model claims someone earns $\leq 50K$, it is correct in 76% of the cases.
- **Recall = 1.00 (100%):** The model managed to find 100% of all actual $\leq 50K$ individuals because it blindly classifies every single person into this category.
- **F1-score = 0.86:** A neat combined score for this large majority group.

The Group $> 50K$ (People earning more than \$50,000)

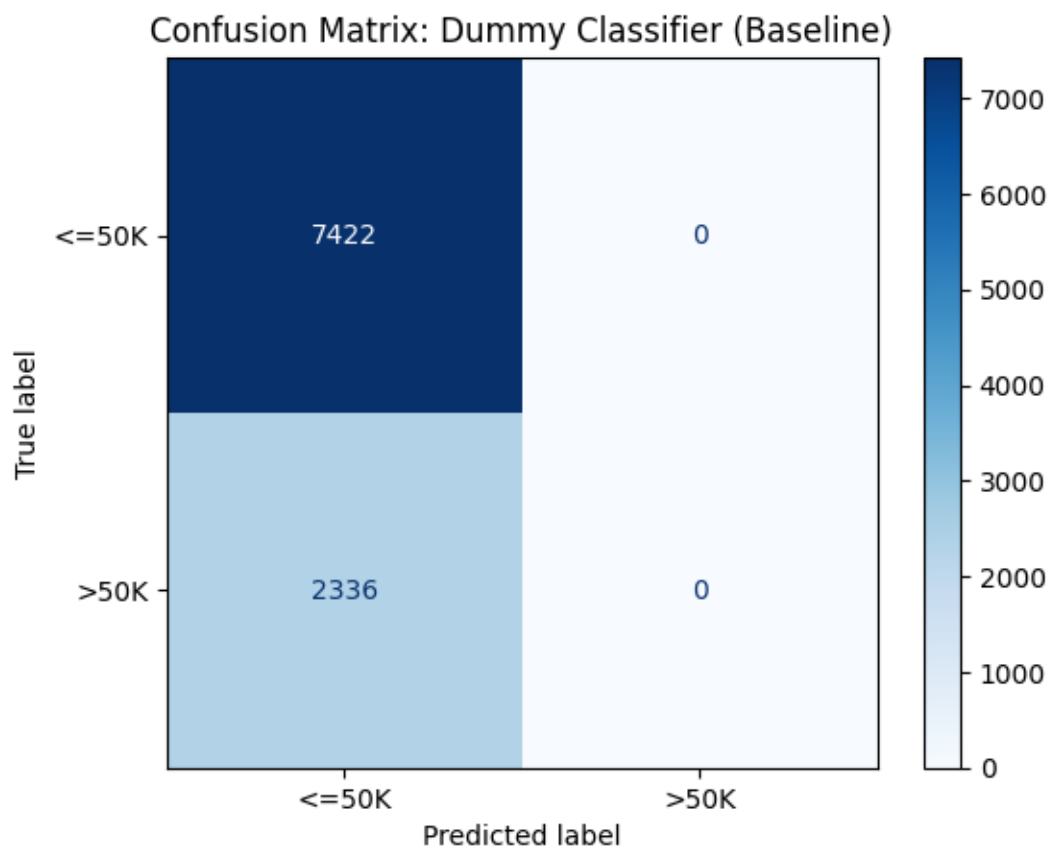
- **Support = 2336:** The total number of wealthier individuals in the test set.
- **Precision = 0.00 (0%):** The model has 0% precision because it never predicts that anyone belongs to the higher-income class.
- **Recall = 0.00 (0%):** The model identified 0% of all actual wealthier people (and missed 100% of them).
- **F1-score = 0.00:** A score of zero, showing the model is completely ineffective for this group.

The Overall Score

- **Accuracy = 0.76 (76%):** Across the board, this model achieved an overall accuracy of 76%.

What is the conclusion about Dummy Classifier (Baseline)?

This baseline model highlights the strong class imbalance in the dataset. While its overall accuracy appears decent at 76%, it achieves this purely by predicting the majority class ($\leq 50K$) for every instance. It completely fails to detect any individuals earning $> 50K$, proving why more advanced models like KNN, Logistic Regression, and Random Forest are necessary.



1. The Confusion Matrix of KNN

The matrix shows that this model actually attempts to recognize both groups:

- **Top-left (6742):** These are people who *actually* earned $\leq 50K$, and the model **correctly** predicted this.
- **Top-right (680):** These are people who earned $\leq 50K$, but the model mistakenly thought they were wealthy ($> 50K$).
- **Bottom-left (927):** These are people who *actually* earned $> 50K$, but the model thought they earned $\leq 50K$. Here, the model missed a portion of the wealthier group.
- **Bottom-right (1409):** These are people who *actually* earned $> 50K$, and the model **correctly** predicted this. This is a massive improvement from the 0 of the previous model!

2. The Classification Report

The Group $\leq 50K$ (People earning \$50,000 or less)

- **Support = 7422:** The total number of people in this actual category.
- **Precision = 0.88 (88%):** When the model claims someone earns $\leq 50K$, it is correct in **88%** of the cases.
- **Recall = 0.91 (91%):** The model managed to track down **91%** of all actual $\leq 50K$ individuals.
- **F1-score = 0.89:** A strong combined score for this group.

The Group $> 50K$ (People earning more than \$50,000)

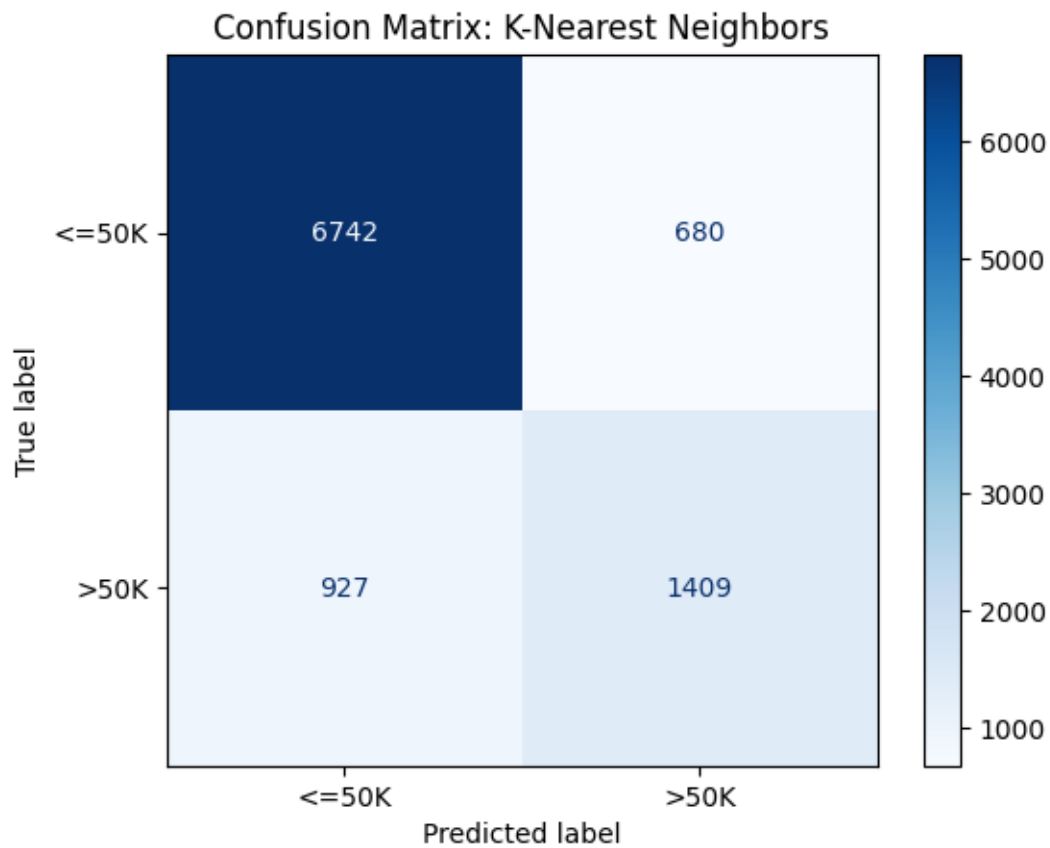
- **Support = 2336:** The total number of wealthier individuals in the test set.
- **Precision = 0.67 (67%):** When the model says someone earns more than 50k, it is correct in **67%** of the cases.
- **Recall = 0.60 (60%):** The model managed to find **60%** of all actual wealthier people (and missed 40%).
- **F1-score = 0.64:** A significant improvement compared to the 0.00 of the previous model.

The Overall Score

- **Accuracy = 0.84 (84%):** Across the board, this KNN model got **84%** of all predictions right.

What is the conclusion about KNN?

This model performs **much better** than the baseline! Where the previous model completely ignored the wealthy group, KNN succeeds in correctly identifying 1409 wealthier people (a recall of 0.60). It is not yet perfect, but it proves that the model is actually extracting patterns from the data instead of just blindly guessing.



1. The Confusion Matrix of Logistic Regression

The matrix illustrates how the model's predictions align with reality:

- **Top-left (6833):** These are people who *actually* earned $\leq 50K$, and the model **correctly** predicted this.
- **Top-right (589):** These are people who earned $\leq 50K$, but the model mistakenly classified them as wealthy ($> 50K$).
- **Bottom-left (973):** These are people who *actually* earned $> 50K$, but the model incorrectly predicted they earned $\leq 50K$.
- **Bottom-right (1363):** These are people who *actually* earned $> 50K$, and the model **correctly** predicted this.

2. The Classification Report

The Group $\leq 50K$ (People earning \$50,000 or less)

- **Support = 7422:** The total number of people in this actual category.
- **Precision = 0.88 (88%):** When the model claims someone earns $\leq 50K$, it is correct in **88%** of the cases.
- **Recall = 0.92 (92%):** The model managed to find **92%** of all actual $\leq 50K$ individuals.
- **F1-score = 0.90:** An excellent combined score for this large group.

The Group $> 50K$ (People earning more than \$50,000)

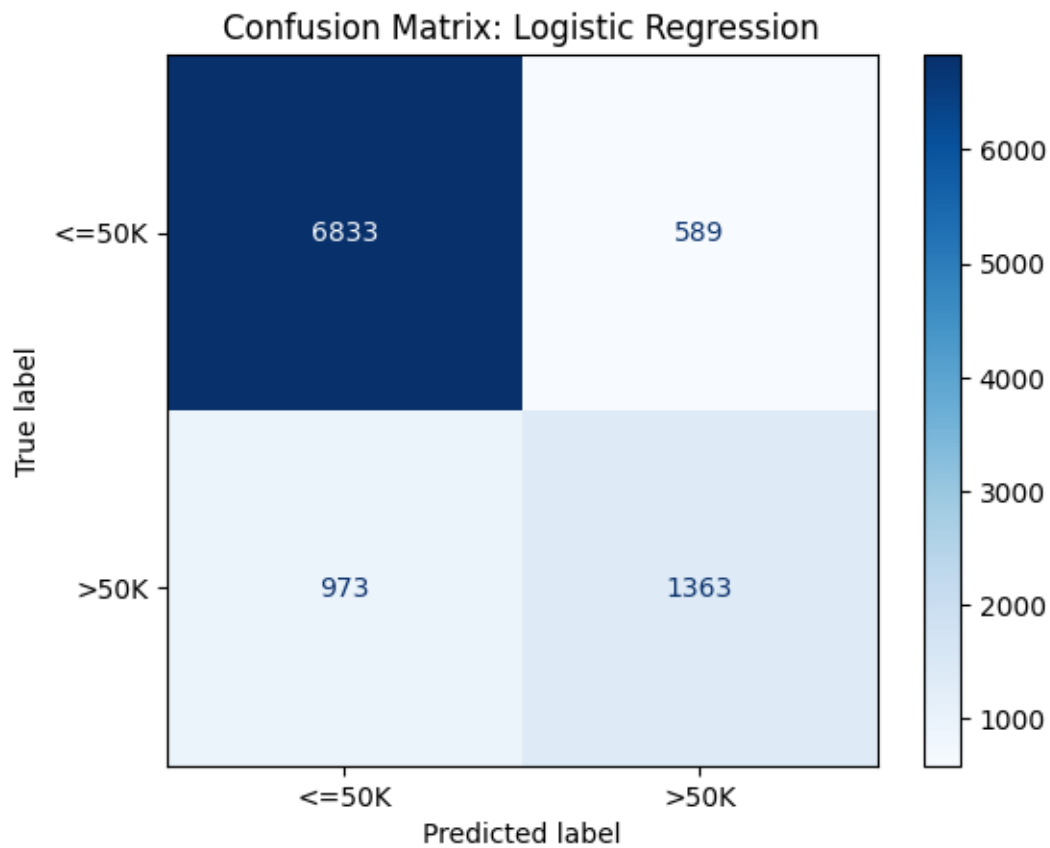
- **Support = 2336:** The total number of wealthier individuals in the test set.
- **Precision = 0.70 (70%):** When the model says someone earns more than 50k, it is correct in **70%** of the cases. This is slightly higher than KNN (67%), meaning it produces fewer false alarms for the higher-income class.
- **Recall = 0.58 (58%):** The model identified **58%** of all actual wealthier people (and missed 42%).
- **F1-score = 0.64:** A stable score, identical to the KNN model.

The Overall Score

- **Accuracy = 0.84 (84%):** Across the board, this model achieved an overall accuracy of **84%**, matching KNN.

What is the conclusion about Logistic Regression?

This model performs almost identically to KNN in overall accuracy (84%). The main advantage of Logistic Regression is its higher **precision** for the higher-income class (70%). This means that when it predicts someone earns $> 50K$, you can be slightly more confident in that prediction compared to KNN.



1. The Confusion Matrix of Random Forest

The matrix shows how the predictions of the Random Forest model turned out:

- **Top-left (6779):** These are people who *actually* earned $\leq 50K$, and the model **correctly** predicted this.
- **Top-right (643):** These are people who earned $\leq 50K$, but the model mistakenly predicted they earned more than 50k ($> 50K$).
- **Bottom-left (892):** These are people who *actually* earned $> 50K$, but the model incorrectly predicted they earned $\leq 50K$.
- **Bottom-right (1444):** These are people who *actually* earned $> 50K$, and the model **correctly** predicted this. This is the highest number of correctly identified higher incomes across all models!

2. The Classification Report

The Group $\leq 50K$ (People earning \$50,000 or less)

- **Support = 7422:** The total number of people in this actual category.
- **Precision = 0.88 (88%):** When the model claims someone earns $\leq 50K$, it is correct in **88%** of the cases.
- **Recall = 0.91 (91%):** The model managed to find **91%** of all actual $\leq 50K$ individuals.
- **F1-score = 0.90:** A very strong and stable score for this group.

The Group $> 50K$ (People earning more than \$50,000)

- **Support = 2336:** The total number of wealthier individuals in the test set.
- **Precision = 0.69 (69%):** When the model says someone earns more than 50k, it is correct in **69%** of the cases.
- **Recall = 0.62 (62%):** The model managed to track down **62%** of all actual wealthier people. This outperforms both KNN (60%) and Logistic Regression (58%).
- **F1-score = 0.65:** The highest F1-score for the higher-income group among all tested models!

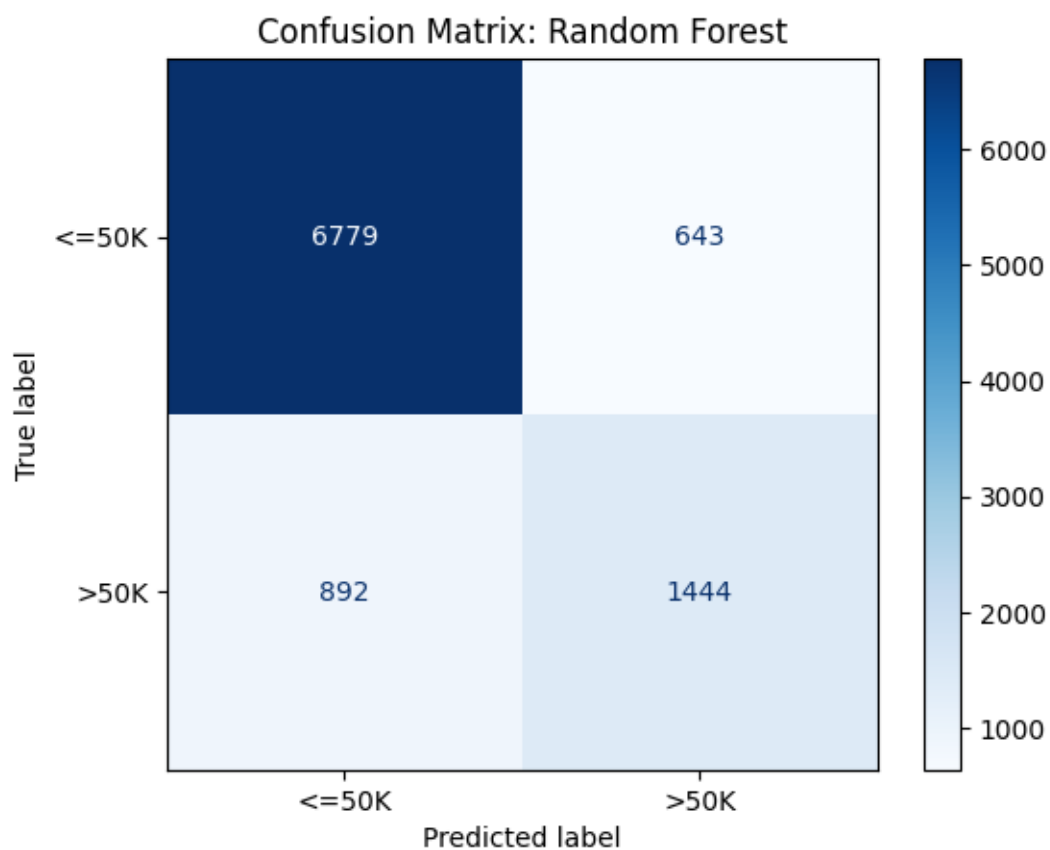
The Overall Score

- **Accuracy = 0.84 (84%):** Overall, the Random Forest model achieved an accuracy of **84%**.

What is the conclusion about Random Forest?

Overall, this is the **best-performing model** in our research!

Although its overall accuracy (84%) matches KNN and Logistic Regression, Random Forest performs significantly better on **recall (62%)** and **F1-score (0.65)** for the higher-income group ($> 50K$). It successfully identifies the largest number of high-income individuals (1,444 people) without making too many false positive errors. That is why choosing this model to make predictions for the fictional person was the right call!



Overall Model Comparison & Conclusion

Across our experimentation, we evaluated four distinct classification models to predict whether a U.S. adult earns at most \$50,000 per year:

1. **Dummy Classifier (Baseline):** Achieved an accuracy of **76%** purely by predicting the majority class ($\leq 50K$) for all instances. It completely failed to identify any individuals earning $> 50K$ (Recall = 0.00, F1-score = 0.00).
2. **K-Nearest Neighbors (KNN):** Improved overall accuracy to **84%** and successfully identified 1,409 high-income individuals, achieving an F1-score of **0.64** for the $> 50K$ class.
3. **Logistic Regression:** Matched KNN's overall accuracy of **84%** and $> 50K$ F1-score of **0.64**, but offered a higher precision of **70%** for the $> 50K$ class, resulting in fewer false positive errors.
4. **Random Forest (Best Model):** Emerged as the strongest model overall. While maintaining an **84%** overall accuracy, it achieved the highest recall (**62%**) and F1-score (**0.65**) for the $> 50K$ income class, correctly detecting 1,444 high earners.

Interpretation of the New Employee Prediction

Using our top-performing **Random Forest** pipeline, we generated a prediction for a 35-year-old male employee working 50 hours a week in an executive-managerial role with a Bachelor's degree.

- **Model Output:** Prediction for new employee: $\leq 50K$ (with 62.1% certainty)
- **Key Insight:** Despite having features often correlated with higher earnings (a Bachelor's degree, 50 weekly work hours, and an executive role), the model predicts an income of $\leq 50K$ with **62.1% probability**.
- **Uncertainty Analysis:** A probability score of 62.1% indicates moderate confidence—it sits relatively close to the 50% decision boundary. This reflects that the employee possesses attributes from both income brackets, illustrating that income levels in the dataset are complex and influenced by a combination of multiple features rather than single job or education titles alone.